

Question	Answer	Marks	Guidance
9(a)	wavelength associated with a <u>moving</u> particle	B1	Ignore any named particles.
9(b)	wave nature (matter): electron diffraction / interference	B1	Allow diffraction of protons, neutrons or any named nuclide.
	particle nature (matter): e.g. states of matter / state changes / gas behaviour / quantisation of charge / Kirchhoff's first law / electrostatics / radioactive decay / properties of ionising radiation / Rutherford scattering / matter–antimatter annihilation	B1	Allow 'Brownian motion'. Allow 'pair' for 'matter – antimatter'.
	wave nature (EM radiation): diffraction / interference / polarisation in light / Doppler effect in light	B1	Allow 'refraction in light'. Allow 'redshift' for 'Doppler effect'. Allow any <u>named</u> part of the EM spectrum for 'light'. Allow 'Young's slits experiment' without the need to qualify as 'with light'.
	particle nature (EM radiation): photoelectric effect	B1	Allow 'photoelectric emission' or 'photoemission'. Allow emission / absorption spectra. Allow 'Compton scattering'.
9(c)(i)	$qV = \frac{1}{2}mv^2$	C1	
	$1.60 \times 10^{-19} \times 3500 = \frac{1}{2} \times 9.11 \times 10^{-31} \times v^2$	C1	
	speed = $3.5 \times 10^7 \text{ m s}^{-1}$	A1	Correct to at least 2 significant figures. AFC. (3.51)
9(c)(ii)	wavelength = h / mv	C1	Allow $\lambda = h / p$ and $p = mv$.
	$= (6.63 \times 10^{-34}) / (9.11 \times 10^{-31} \times 3.5 \times 10^7)$ $= 2.1 \times 10^{-11} \text{ m}$	A1	Possible ECF from 9(c)(i) . Correct to at least 2 significant figures. AFC.